

RecAL Digital Product Passport: Enhancing Traceability and Transparency in the **Al**uminum Supply Chain

A DLT-Backed Solution for EU Circular Economy Compliance

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Project Consortium

19 Partners
8 EU Countries
+ UK



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Recycling Hub for Aluminium



Platform



New Alloys

Optimising the impurity tolerance of next-generation alloy designs



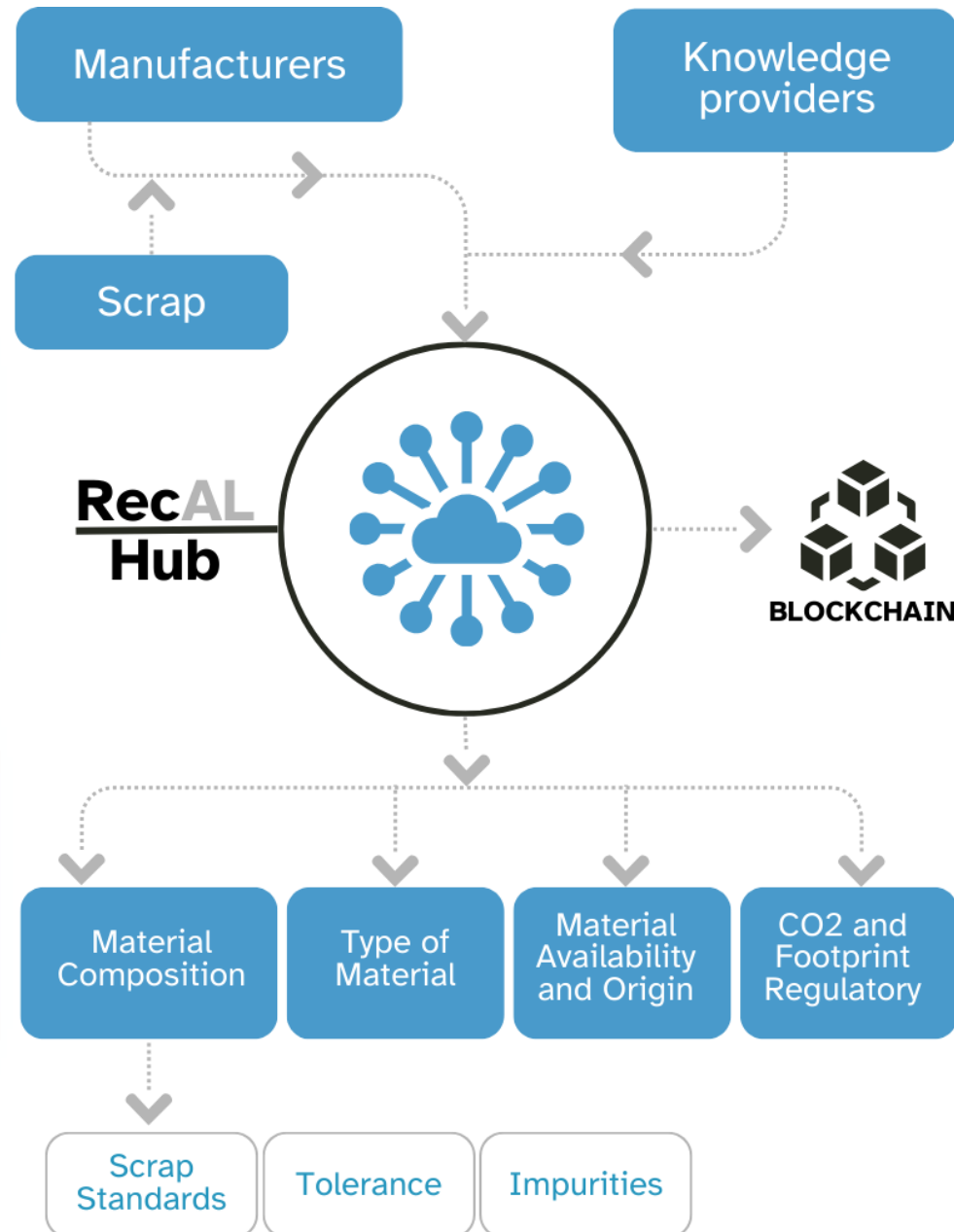
Waste purity

Optimization of waste purity through advanced sorting & refining technologies for EoL aluminium



DPP & Data

Promoting the circular economy by providing data for the carbon footprint & digital product passport (DPP) via blockchain



Motivation

Why Digital Product Passports (DPPs)?



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Policy Landscape: EU Green Deal & ESPR

- ✓ The **Ecodesign for Sustainable Products Regulation (ESPR)** mandates DPPs for priority sectors
- ✓ A core pillar of the **EU Circular Economy Action Plan**, aiming to make sustainable products the norm.
- ✓ DPPs will be **mandatory** for aluminum products



Industry Challenges: Why Aluminum Needs DPPs

- ✓ **Low traceability** in raw material sourcing (e.g., bauxite origins).
- ✓ **High carbon footprint** from energy-intensive smelting processes.
- ✓ **Recycled content fraud** undermines sustainability claims.
- ✓ **Lack of real-time data** for regulators, manufacturers, and consumers.



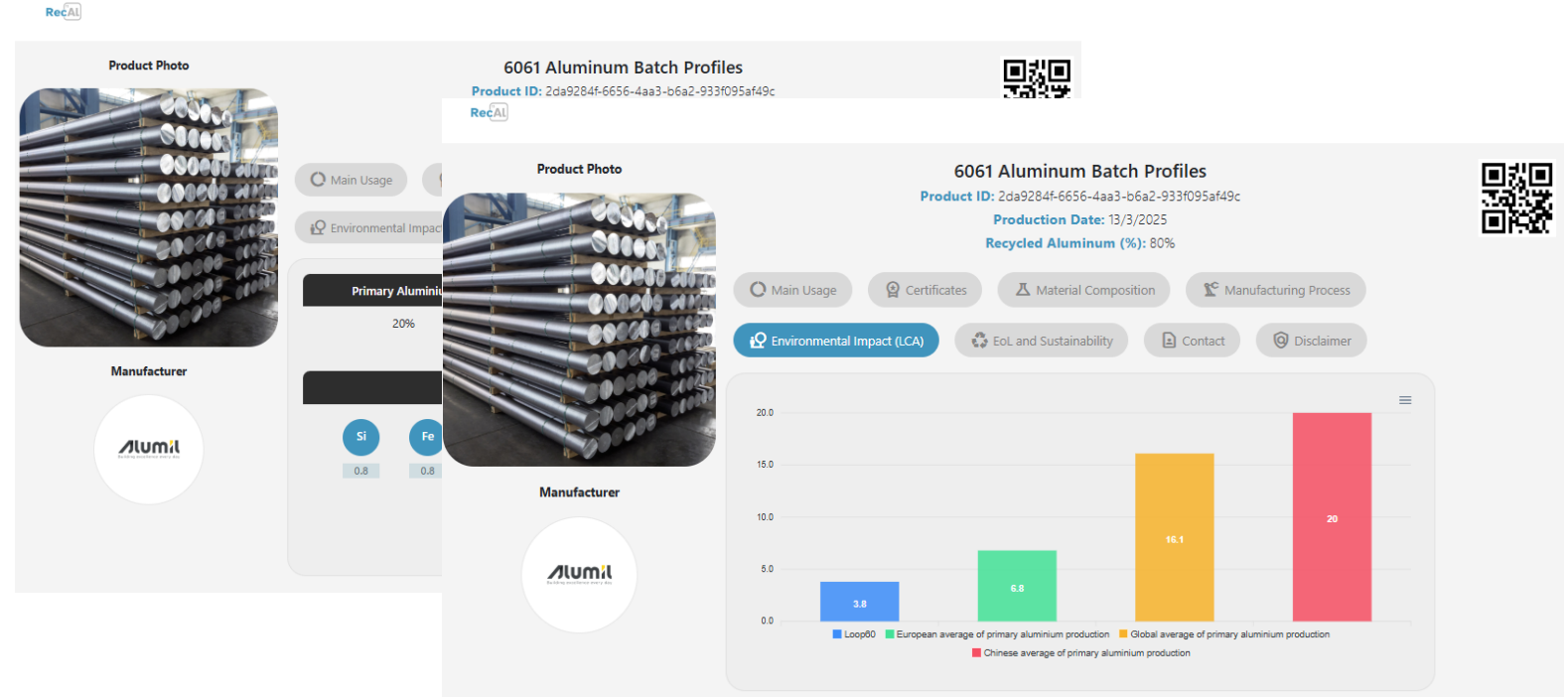
The DPP Opportunity

- ✓ A DPP acts as a **digital twin** for each aluminum product.
- ✓ Provides **structured, verifiable data** across the product lifecycle:
- ✓ Empowers stakeholders to make **data-driven decisions**.
- ✓ Creates **transparency and trust** in a fragmented, globalized supply chain.

RecAL DPP Overview

What is the RecAL DPP?

- **Governance:** Hybrid decentralized model (RecAL Hub + blockchain).
- **Domain:** Blockchain-secured digital record for aluminum batches.
- **Key Data:** Material composition, carbon footprint, certifications, EoL status mapped to ESPR requirements
- **Access:** QR codes for stakeholders (manufacturers, recyclers, regulators).



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Why Blockchain + DPP

Key Benefits for Aluminum Industry



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Ethical Sourcing

- Track origin of bauxite
- Certify conflict-free & ESG-compliant materials
- Supplier accountability on-chain



Carbon Footprint & Green Labels

- Record energy mix (e.g., hydro vs. coal)
- Automate green certifications (e.g., CBAM-ready)
- Trace full lifecycle emissions



Circular Economy

- Unique IDs for recycled Al
- Track alloy type & scrap reuse history
- Enable smart buy-back programs



Anti-Counterfeiting & Quality Assurance

- Embed certifications & test reports
- Use IoT for secure shipping tracking
- Trigger real-time recalls via smart contracts



Simplified Compliance & Trade

- Auto-generate docs: origin, safety, CBAM
- Unified access for all supply chain actors
- Conflict mineral compliance by design

Technology Stack

How It Works: DLT & Smart Contracts



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DLT Backbone

- Built on **Hyperledger Fabric** (permissioned blockchain)
- Participants are known & trusted — ideal for industry



Smart Contracts (Chaincode)

- Encodes compliance logic (e.g., recycled content checks)
- Immutable, distributed business rules



DPP REST API

- Gateway to blockchain services
- Supports secure **CRUD & query operations**
- Integrates with external systems via standard interface



RecAL HUB

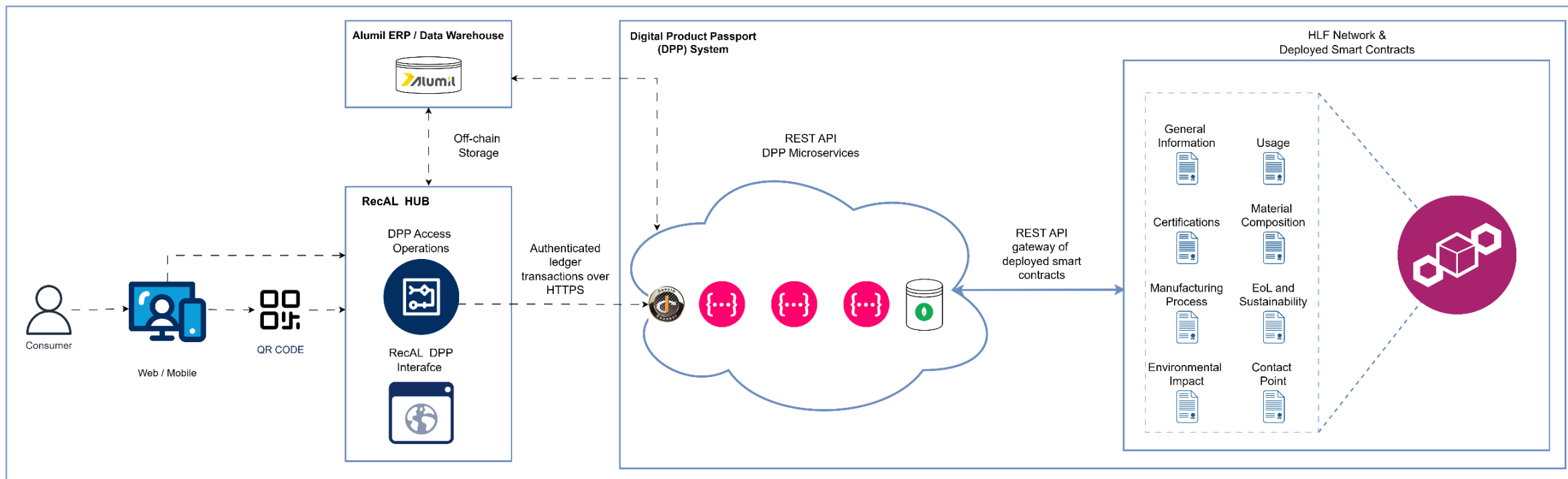
- Federated **data mesh** across supply chain actors
- Interfaces with ERPs, warehouses, and legacy systems
- Enforces **data sovereignty & interoperability**



Semantic Data Model

- Ensures standardization & cross-platform compatibility
- Derived from platform-independent DPP vocabulary

System Architecture



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The HUB

The screenshot displays the RecAL Hub interface. At the top, it says "welcome to your RecAL Hub". Below this, there's a "RecAL Suit" section with seven icons: Data Sheets & Traceability, Data Insights & Missing Data, RecAL Life Cycle Assessment, RecAL Shop, Digital Product Passport (highlighted with a red box), Find new Alloys Research, and Request Alloys R&D. To the left is a sidebar with navigation options: Home, Data, Insights, LCA, Shop, and DPP. Below the RecAL Suit section, there's a section titled "Your aluminium circularity report Q4/2026" which contains a circular flow diagram of the aluminium lifecycle. The diagram consists of eight numbered steps: 1. PRODUCT SOURCING, 2. ASSEMBLY, 3. UTILISATION, 4. COLLECTING & DISMANTLING, 5. SHREDDING, 6. SORTING, 7. MANUFACTURING, and 8. FORMING / CASTING. Arrows indicate the flow from step 1 through to step 8, with a feedback loop from step 4 back to step 1. A recycling symbol is also present in the diagram. To the right of the diagram is a 3D isometric illustration of an industrial facility with a large recycling symbol in the center.



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The HUB

Alumil

Your Digital Data Sheets

Choose your product group

Systems

Frames

Billets

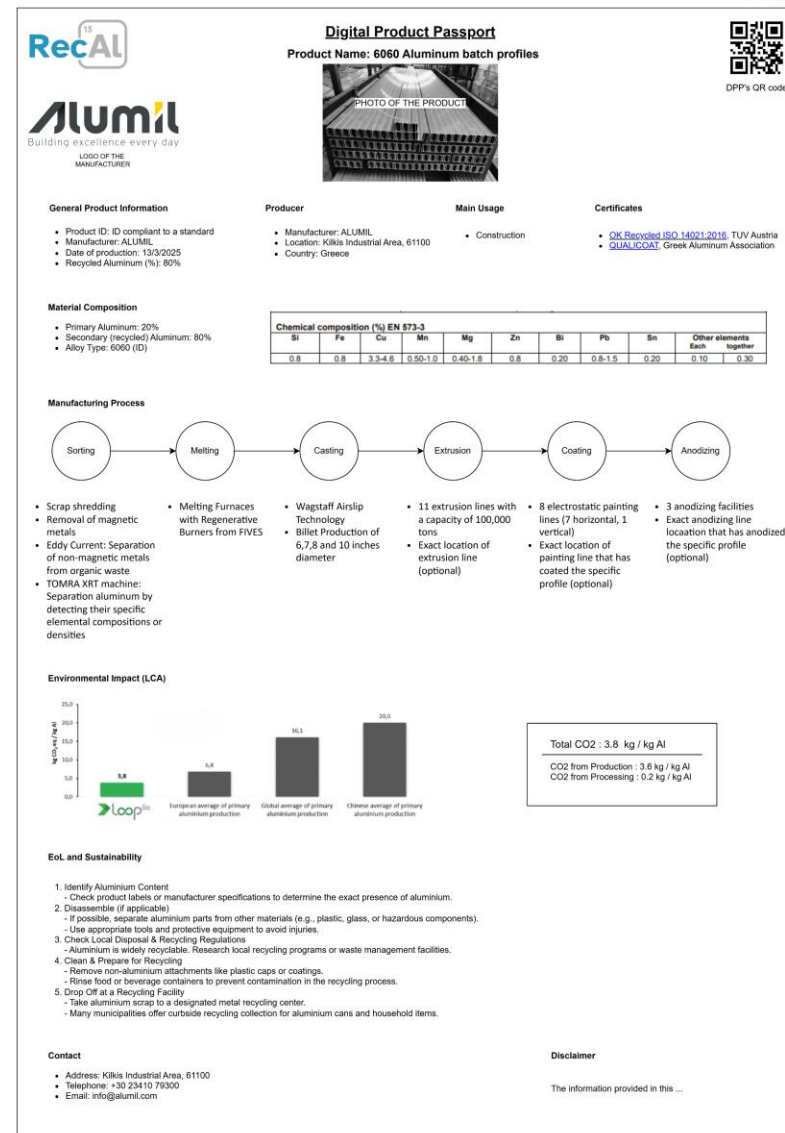
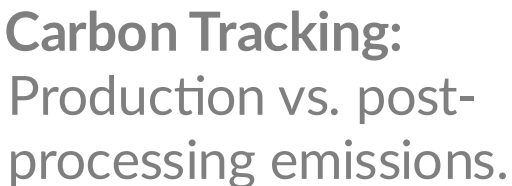
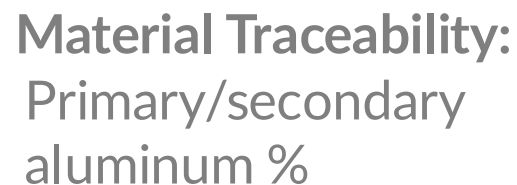
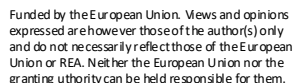
L-Shapes

Search by batch number / designation / product name / material / ...



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Core DPP Capabilities, from Compliance to Value Creation



Benefits

Why This Matters



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For Industry:

Fraud prevention, streamlined reporting, ESG compliance



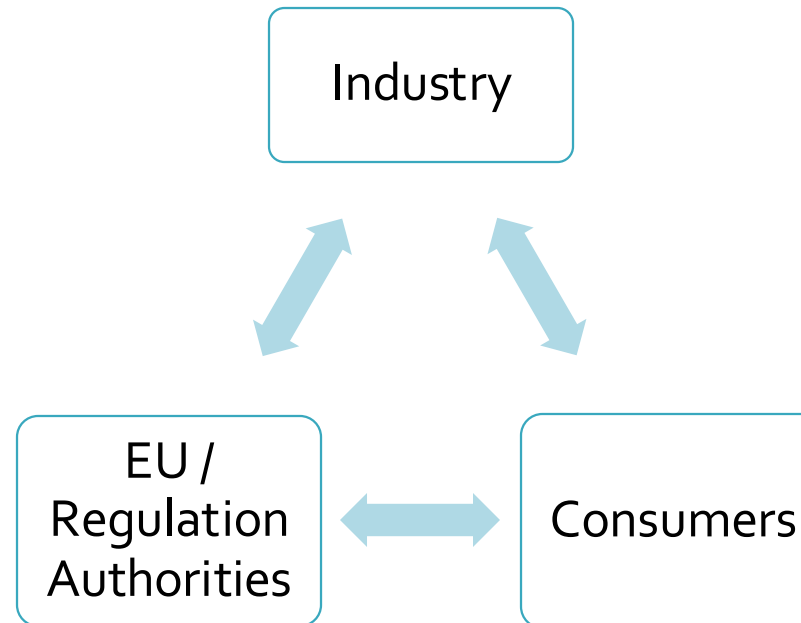
For EU and Policymakers:

ESPR alignment, circular economy goals



For Consumers:

Trust via transparent sustainability data



Challenges & Future Work

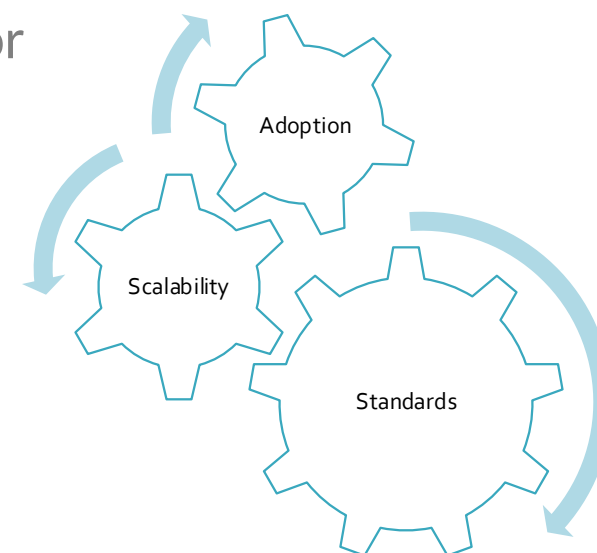
✚ **Challenge:** Integration with Industries Legacy systems

📐 **Data Standardization:** Common schema for interoperability

🚀 **Scalability:** Hybrid on/off-chain storage (RecAL Hub)

🧪 **Adoption:** Pilot testing at ALUMIL

🔄 **Extension:** Adaptable framework for other industries.



Thank you!

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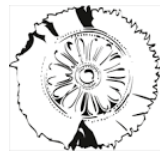
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